

Lecture 8 – Rotational motion of rigid bodies

- A/Prof Tay Seng Chuan

Email: pvotaysc@nus.edu.sg

In this lecture we consider the motion of a **rigid body** rotating about a fixed axis. It will be useful to consider the **rigid body** as being made up of a large number of particles with a total mass m . The **body** is **rigid** because each of these particles is fixed with respect to every other constituent particles.

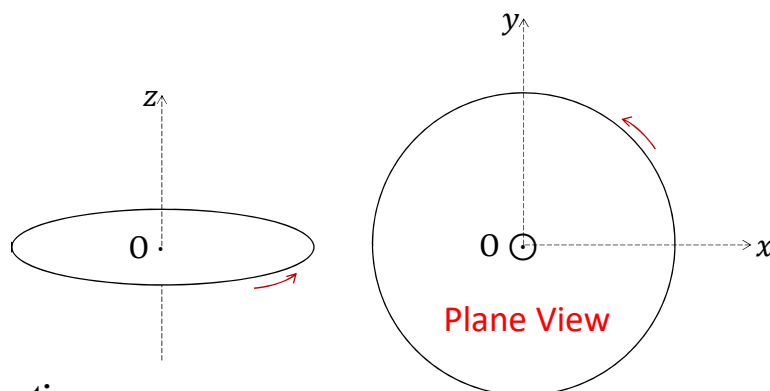


A few slides in this chapter come from Chapter-9 of Young's textbook. This PPT file will well define the scope of your tests.

1

Angular position, angular velocity, and angular acceleration

Consider a **rigid body** that rotates about a fixed axis – the z-axis:



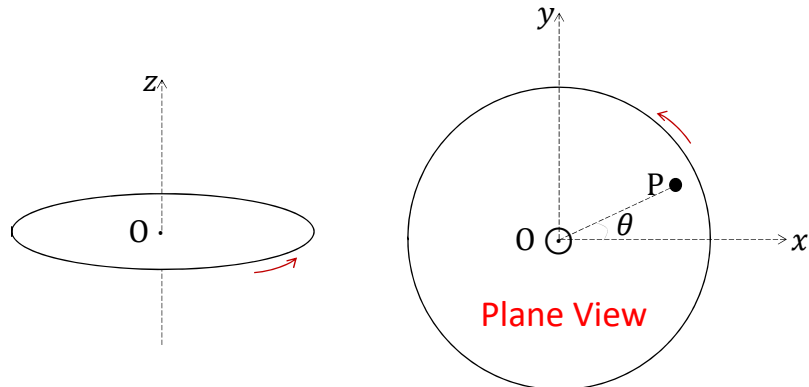
Question.

How does one describe the **rotational motion** of such a **rigid body** as there isn't any translational displacement?

2

Angular position, angular velocity, and angular acceleration (cont.)

Consider a particle P , with mass m , of the **rigid body**:



Suppose the (**perpendicular**) **distance** from O to P is r and the line OP makes an angle θ with the x -axis, the x - and y -coordinates of P are given by $x = r \cos \theta$ and $y = r \sin \theta$.

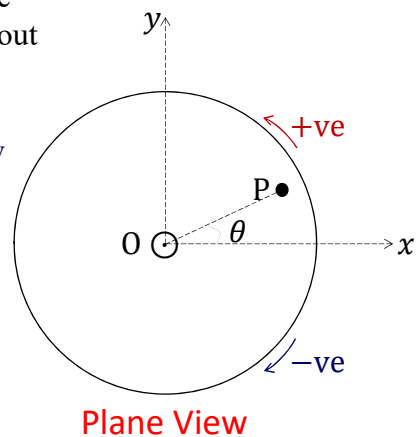
3

Angular position, angular velocity, and angular acceleration (cont.)

Since the **body** is **rigid**, r does not change with time t . As the **rigid body** rotates about the z -axis, only θ changes with time t .

The **angular position** θ of the **rigid body** changes with **time** t .

θ is a **function** of time t : $\theta = \theta(t)$.

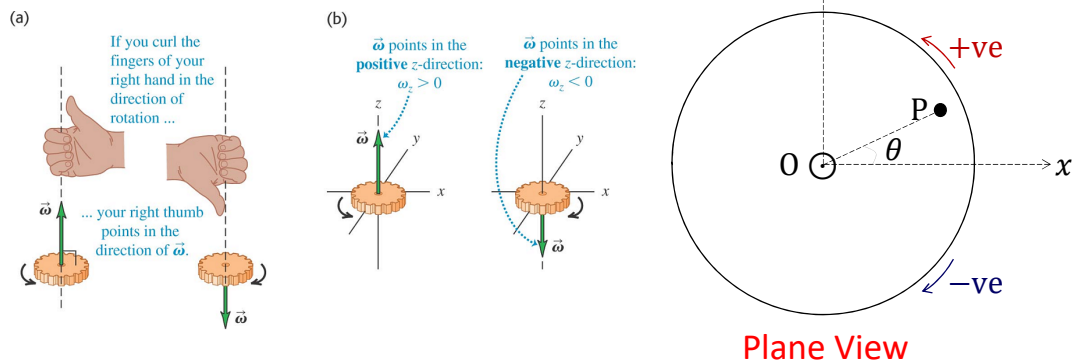


4

Angular position, angular velocity, and angular acceleration (cont.)

The **angular velocity** of the **rigid body**,

$$\omega_z = \omega_z(t) = \frac{d}{dt} \theta(t).$$



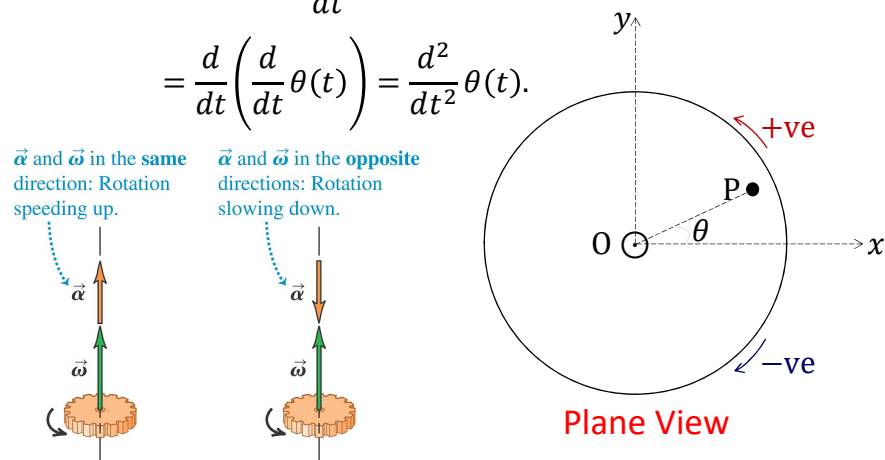
5

Angular position, angular velocity, and angular acceleration (cont.)

The **angular acceleration** of the **rigid body**,

$$\alpha_z = \alpha_z(t) = \frac{d}{dt} \omega_z(t)$$

$$= \frac{d}{dt} \left(\frac{d}{dt} \theta(t) \right) = \frac{d^2}{dt^2} \theta(t).$$



6

Example 1. Rotation about a fixed axis, the z-axis, with constant angular acceleration α_{0z}

Let θ_0 and ω_{0z} be the angular position and angular velocity respectively of a rigid body at time $t = 0$, and let angular acceleration

$\alpha_z = \frac{d\omega_z}{dt}$, the angular velocity at any later time $t > 0$ is

$$\omega_z(t) = \omega_{0z} + \int_0^t \alpha_z(t') dt' = \omega_{0z} + \int_0^t \alpha_{0z} dt' = \omega_{0z} + \alpha_{0z}t$$

angular acceleration has been assumed to be a constant

Since $\omega_z = \frac{d\theta}{dt}$, its angular position at any later time $t > 0$ is

$$\begin{aligned} \theta(t) &= \theta_0 + \int_0^t \omega_z(t') dt' = \theta_0 + \int_0^t (\omega_{0z} + \alpha_{0z}t') dt' \\ &= \theta_0 + \omega_{0z}t + \frac{1}{2}\alpha_{0z}t^2 \end{aligned}$$

7

Example 1. Rotation about a fixed axis, the z-axis, with constant angular acceleration α_{0z}

Translational Kinematics

Rotational Kinematics

$$v_x(t) = v_{0x} + a_{0x}t \quad \rightarrow \quad \omega_z(t) = \omega_{0z} + \alpha_{0z}t$$

$$x(t) = x_0 + \left(v_{0x}t + \frac{1}{2}a_{0x}t^2 \right) \quad \rightarrow \quad \theta(t) = \theta_0 + \left(\omega_{0z}t + \frac{1}{2}\alpha_{0z}t^2 \right)$$

$$x(t) = x_0 + \frac{(v_{0x} + v_x(t))}{2} \times t \quad \rightarrow \quad \theta(t) = \theta_0 + \frac{(\omega_{0z} + \omega_z(t))}{2} \times t$$

$$v_x^2 = v_{0x}^2 + 2a_{0x}(x - x_0) \quad \rightarrow \quad \omega_z^2 = \omega_{0z}^2 + 2\alpha_{0z}(\theta - \theta_0)$$

8

Example 1. Rotation about a fixed axis, the z-axis, with constant angular acceleration α_{0z}

Translational Kinematics

Rotational Kinematics

Simplified Version

$$v_f = v_i + at \rightarrow \omega_f = \omega_i + \alpha t$$

$$\Delta x = v_i t + \frac{1}{2} at^2 \rightarrow \Delta \theta = \omega_i t + \frac{1}{2} \alpha t^2$$

$$\Delta x = \frac{(v_i + v_f)}{2} \times t \rightarrow \Delta \theta = \frac{(\omega_i + \omega_f)}{2} \times t$$

$$v_f^2 = v_i^2 + 2a\Delta x \rightarrow \omega_f^2 = \omega_i^2 + 2\alpha\Delta \theta$$

Is there any difference?

9

Example 1. Rotation about a fixed axis, the z-axis, with constant angular acceleration α_{0z}

Translational Kinematics

Rotational Kinematics

Simplified Version

$$v_f = v_i + at \rightarrow \omega_f = \omega_i + \alpha t$$

$$\Delta x = v_i t + \frac{1}{2} at^2 \rightarrow \Delta \theta = \omega_i t + \frac{1}{2} \alpha t^2$$

$$\Delta x = \frac{(v_i + v_f)}{2} \times t \rightarrow \Delta \theta = \frac{(\omega_i + \omega_f)}{2} \times t$$

$$v_f^2 = v_i^2 + 2a\Delta x \rightarrow \omega_f^2 = \omega_i^2 + 2\alpha\Delta \theta$$

Is there any difference?

9

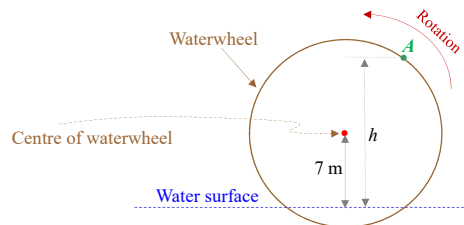
Consider both you and I are running along a straight line, and I am ahead of you. Both of us maintain 2 different constant speeds and you are faster. In this scenario you will overtake me once. As our separation will become farther and farther, there is no way for both of us to meet again.

But we will meet many times if we are running around a circle!!

10

Problem. A waterwheel rotates evenly at 6 revolutions per minute. **Three seconds** after Tom starts a stopwatch, point A on the rim of the wheel is at its highest possible altitude (i.e., h , as measured from the water surface, has a value of **15 m** at that highest possible moment) above the surface of the water. The centre of the waterwheel is 7 m above the water surface.

- How high above or below the surface of the water will point A be at time 17.5 s on the stopwatch of Tom?
- At what time on the stopwatch of Tom was point A first emerging from the water, and subsequently at what time did it enter the water?



Answer:

11

Problem. A waterwheel rotates evenly at 6 revolutions per minute. **Three seconds** after Tom starts a stopwatch, point A on the rim of the wheel is at its highest possible altitude (i.e., h , as measured from the water surface, has a value of **15 m** at that highest possible moment) above the surface of the water. The centre of the waterwheel is 7 m above the water surface.

- How high above or below the surface of the water will point A be at time 17.5 s on the stopwatch of Tom?
- At what time on the stopwatch of Tom was point A first emerging from the water, and subsequently at what time did it enter the water?

Answer: The problem is that we do not know the angular position of point A when time = 0.

- Let the angular position of point A at time t correspond to the angle θ_t , where θ_t is measured from the horizontal line through the center of the waterwheel and it is measured in counter clockwise direction.

Let θ_0 be the corresponding position angle of point A when $t = 0$. Let ω be the angular speed of the waterwheel in radian per second.

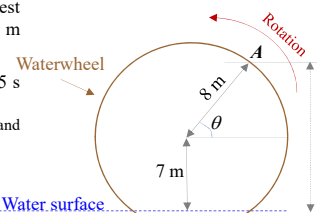
If the waterwheel is rotating at constant angular speed, we will have

$$\theta_t = \theta_0 + \omega t, \text{ where } \omega = \frac{6 \times 2\pi}{60} \text{ rad/sec} = \frac{\pi}{5} \text{ rad/sec.}$$

Give that when $t = 3$, $h = 15$ (highest possible), it also means that when $t = 3$, $\theta_3 = \frac{\pi}{2} + 2n\pi$, where $n = 0, 1, 2, 3, \dots$

For 3 sec, $\Delta\theta = \omega \times t = \frac{\pi}{5} \times 3 = \frac{3\pi}{5} < 2\pi$. So $n = 0$.

$$\theta_3 = \theta_0 + \omega \times 3 \Rightarrow \frac{\pi}{2} = \theta_0 + \frac{\pi}{5} \times 3 \Rightarrow \theta_0 = -\frac{\pi}{10}$$



Let h_t be the vertical height of point A measured from the water level at time t .

From the geometry, radius of waterwheel = 15 m - 7 m = 8 m.

$$h_t = 7 + 8 \sin(\theta_t) \rightarrow h_t = 7 + 8 \sin(\theta_0 + \omega t)$$

When $t = 17.5$ sec,

$$\begin{aligned} h_{17.5} &= 7 + 8 \sin\left(-\frac{\pi}{10} + \frac{\pi}{5}(17.5)\right) \\ &= 7 + 8 \sin(3.4\pi) \\ &= 7 + 8 \times (-0.9511) \\ &= \boxed{-0.6084} \end{aligned}$$

i.e., 0.6 m below the water surface.

12

- (ii) At what time on the stopwatch of Tom was point A first emerging from the water, and subsequently at what time did it enter the water?

Answer:

We let $h_t = 0$ in $h_t = 7 + 8 \sin(\theta_0 + \omega t) = 0$

$$\rightarrow \sin\left(-\frac{\pi}{10} + \frac{\pi}{5}t\right) = -\frac{7}{8}$$

$$-18^\circ + 36^\circ \times t = 241.05^\circ, 601.05^\circ, 961.05^\circ, \dots$$

or $298.96^\circ, 658.96^\circ, 1018.96^\circ, \dots$

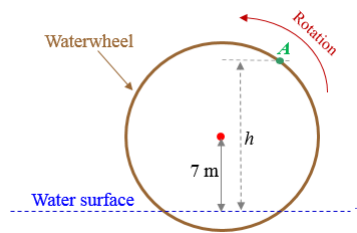
i.e., $-18^\circ + 36^\circ \times t = 241.05^\circ, 298.96^\circ, 601.05^\circ, 658.96^\circ, 961.05^\circ, 1018.96^\circ, \dots$

So $t = 7.2 \text{ s}, 8.8 \text{ s}, 17.2 \text{ s}, 18.8 \text{ s}, 27.2 \text{ s}, 28.8 \text{ s}, \dots$ when the height of point A is 0 (entering the water or leaving the water)

$$\theta_0 = -\frac{\pi}{10} \quad \text{i.e.,} \quad \theta_0 = -\frac{180^\circ}{10} = -18^\circ$$

From the geometry, $\beta = \sin^{-1}\left(-\frac{7}{8}\right) = -61.04^\circ$. So point A was initially **above** the water surface when $t = 0$.

So when $t = 8.8 \text{ s}$ point A first emerged from the water, and subsequently when $t = 17.2 \text{ s}$ it entered the water again.



S	All
✓ T	C ✓

13

Example. A waterwheel rotates evenly at 6 revolutions per minute. Three seconds after Tom starts a stopwatch, point A on the rim of the wheel is at its highest possible altitude (i.e., h , as measured from the water surface, has a value of 15 m at that highest possible moment) above the surface of the water. The centre of the waterwheel is 7 m above the water surface.

- (i) How high above or below the surface of the water will point A be at time 17.5 s on the stopwatch of Tom?
 (ii) At what time on the stopwatch of Tom was point A first emerging from the water, and subsequently at what time did it enter the water?

Answer:

- (i) Let the position of point A at time t correspond to the angle θ_t where θ_t is measured from the horizontal line through the center of the waterwheel and it is measured in counter clockwise direction.

Let θ_0 be the corresponding position angle of point A when $t = 0$. Let ω be the angular speed of the waterwheel in radian per second.

If the waterwheel is rotating at constant angular speed, we will have $\theta_t = \theta_0 + \omega t$, where $\omega = \frac{6 \times 2\pi}{60} \text{ rad/sec} = \frac{\pi}{5} \text{ rad/sec}$.

Give that when $t = 3$, $h = 15$ (m), it also means that when $t = 3$, $\theta_t = \frac{\pi}{2} + 2n\pi$, where $n = 0, 1, 2, 3, \dots$

For 3 sec, $\Delta\theta = \omega \times t = \frac{\pi}{5} \times 3 = \frac{3\pi}{5} < 2\pi$. So $n = 0$, $\theta_3 = \theta_0 + \omega \times 3 \rightarrow \frac{\pi}{2} = \theta_0 + \frac{\pi}{5} \times 3 \rightarrow \theta_0 = -\frac{\pi}{10}$

Let h_t be the vertical height of point A measured from the water level at time t . From the geometry, radius of waterwheel = 15 m - 7 m = 8 m.
 $h_t = 7 + 8 \sin(\theta_t) \rightarrow h_t = 7 + 8 \sin(\theta_0 + \omega t)$

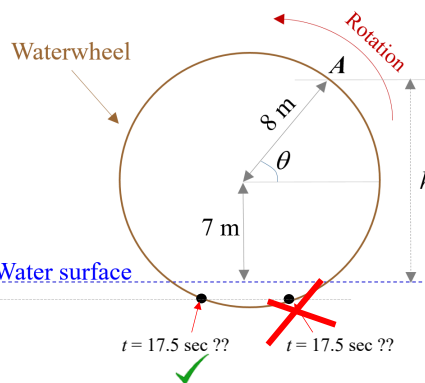
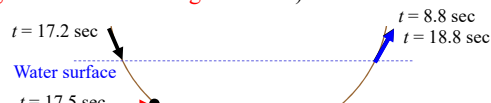
When $t = 17.5$ sec,

$$\begin{aligned} h_{17.5} &= 7 + 8 \sin\left(-\frac{\pi}{10} + \frac{\pi}{5}(17.5)\right) \\ &= 7 + 8 \sin(3.4 \times \pi) \\ &= 7 + 8 \times (-0.9511) \\ &= [-0.60884] \end{aligned}$$

i.e., 0.6 m below the water surface.

First emerge from water surface

From (ii): So $t = 7.2 \text{ s}, 8.8 \text{ s}, 17.2 \text{ s}, 18.8 \text{ s}, 27.2 \text{ s}, 28.8 \text{ s}, \dots$ when the height of point A is 0 (entering the water or leaving the water)



$$t = \frac{17.2 + 18.8}{2} = 18 \text{ sec}$$

14

Example 1. Rotation about a fixed axis, the z -axis, with constant angular acceleration α_{0z}

If you have finished watching a movie on DVD and the disc is slowing to a stop. The disc's **angular acceleration** is a **constant** $\alpha_{0z} = -10.0 \text{ rad/s}^2$.

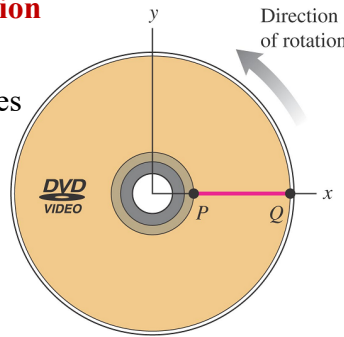
At time $t = 0$, a line PQ on the disc's surface lies along the x -axis, i.e., $\theta_0 = 0$, and its angular velocity is $\omega_{0z} = 27.5 \text{ rad/s}$. At time $t = 0.300 \text{ s}$, the disc's angular velocity,

$$\begin{aligned}\omega_z &= \omega_{0z} + \alpha_{0z}t \\ &= 27.5 - 10 \times 0.3 = 24.5 \text{ rad/s}.\end{aligned}$$

Its angular position,

$$\theta = \theta_0 + \omega_{0z}t + \frac{1}{2}\alpha_{0z}t^2 = 0 + 27.5 \times 0.3 + \frac{1}{2} \times (-10) \times 0.3^2 = 7.8 \text{ rad}.$$

At time $t = 0.300 \text{ s}$, the line PQ makes an angle $(7.8 - 2\pi) = 1.52 \text{ rad}$ with the x -axis.

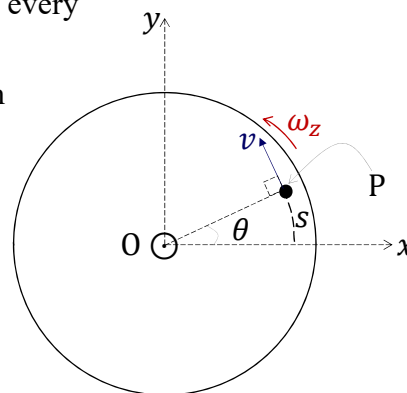


15

Angular position, angular velocity, and angular acceleration (cont.)

Remarks.

- When a **rigid body** rotates about a fixed axis, every particle in the body moves in a circular path.
- For example, P is a (constant) distance r from the axis of rotation, so it moves in a circle of radius r – the circle lies in a plane perpendicular to the axis and centred on the axis.
- At any time t the angle θ and the arc length s are related by $\theta = \frac{s}{r}$ rad. So, $s = r\theta$, where $2\pi \text{ rad} = 360^\circ$, or $1 \text{ rad} = \frac{360^\circ}{2\pi} = 57.3^\circ$.



16

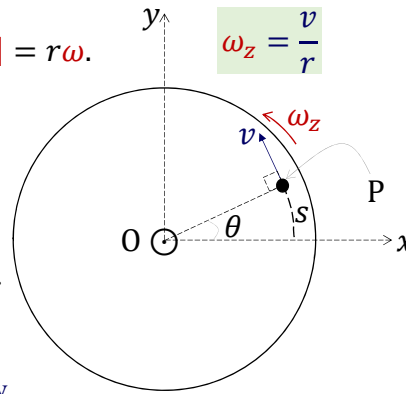
Angular position, angular velocity, and angular acceleration (cont.)

Remarks (cont.).

- The **linear speed (or tangential speed)** of P,

$$v = \left| \frac{ds}{dt} \right| = \left| \frac{r(d\theta)}{dt} \right| = r \frac{d\theta}{dt} = r|\omega_z| = r\omega.$$

- At any instant, every part of a rotating **rigid body** has the same **angular velocity** ω_z – all points rotate through the same angle $d\theta$ in the same time interval dt since the **body** is **rigid**.
- However, depending on how far each point lies from the rotation axis, i.e., r , different points on a rotating **rigid body** move **different distances** ds in a given time interval dt .



17

Angular position, angular velocity, and angular acceleration (cont.)

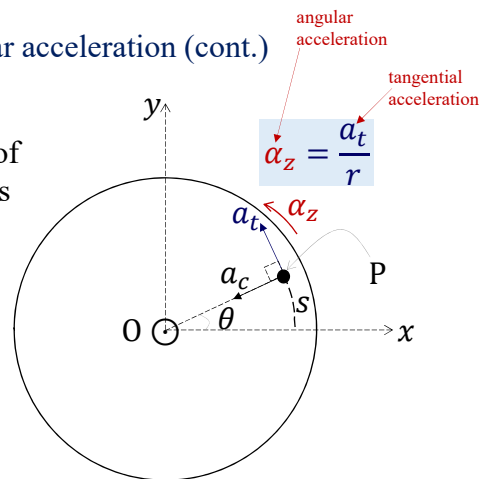
Remarks (cont.).

- $\theta = \theta(t)$ need not be a linear function of time t , i.e., $\theta \neq \theta_0 + \omega_{0z}t$ where ω_{0z} is some constant,

and $\frac{d^2\theta}{dt^2}$ may not be equal to 0.

- The **tangential component of the acceleration** of P,

$$a_t = r \frac{d^2\theta}{dt^2} = r \frac{d\omega_z}{dt} = r\alpha_z, \quad |a_t| = r|\alpha_z| = r\alpha.$$



18

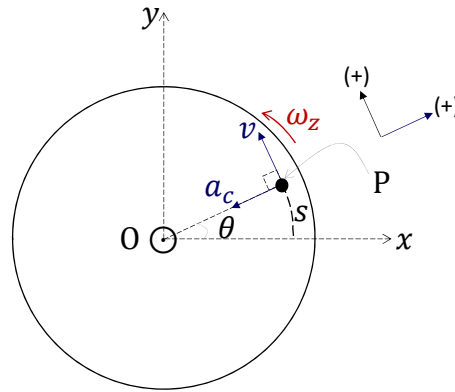
Angular position, angular velocity, and angular acceleration (cont.)

Remarks (cont.).

- The **centripetal component of the acceleration** of P,

$$a_c = -\frac{v^2}{r} = -r\omega^2, \quad |a_c| = r\omega^2.$$

Do not mix up centripetal acceleration with tangential acceleration ($r\alpha$)!!



19

Question. A computer disk drive is turned on starting from rest and has constant angular acceleration. If it took 0.610 s for the drive to make its second complete revolution,

(i) what is its angular acceleration in rad/s²?

**Answer:**

Since the angular acceleration is constant, rotational kinematic formulae can be applied.

Let t_1 be the time taken for the first round, and ω_f be the angular speed after t_1 .

From $\Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2$ for the first round, $2\pi = 0 \times t_1 + \frac{1}{2}\alpha t_1^2 \dots (1)$

From $\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$ for the first round, $\omega_f^2 = 0^2 + 2\alpha(2\pi) = 4\pi\alpha \dots (2)$

Let t_2 be the time taken for the second round, $2\pi = \omega_f t_2 + \frac{1}{2}\alpha t_2^2 \Rightarrow \omega_f = \frac{2\pi - \frac{\alpha t_2^2}{2}}{t_2} \dots (3)$

$$\begin{aligned} \text{Sub (3) in (2), } \left(\frac{2\pi - \frac{\alpha t_2^2}{2}}{t_2} \right)^2 &= 4\pi\alpha \Rightarrow 4\pi^2 - 2(2\pi)\left(\frac{\alpha t_2^2}{2}\right) + \frac{\alpha^2 t_2^4}{4} = 4\pi\alpha t_2^2 \\ &\Rightarrow 4\pi^2 - 6\pi\alpha t_2^2 + \frac{1}{4}\alpha^2 t_2^4 = 0 \end{aligned}$$

20

$$4\pi^2 - 6\pi\alpha t_2^2 + \frac{1}{4}\alpha^2 t_2^4 = 0 \rightarrow 4(3.1416)^2 - 6(3.1416)\alpha(0.61)^2 + \frac{1}{4}\alpha^2(0.61)^4 = 0$$

$$39.47842 - 7.01392\alpha + 0.03461\alpha^2 = 0 \rightarrow 0.03461\alpha^2 - 7.01392\alpha + 39.47842 = 0$$

$$\alpha = \frac{-(-7.01392) \pm \sqrt{(-7.01392)^2 - 4(0.03461)(39.47842)}}{2(0.03461)}$$

$$= \frac{7.01392 \pm \sqrt{43.72978}}{0.06922} = \frac{7.01392 \pm 6.61285}{0.06922} \rightarrow \alpha = 5.79 \text{ rad/s}^2 \text{ or } 196.86 \text{ rad/s}^2$$

It means the disk started from rotating in the opposite direction!! But the disk started from rest.



$$\omega_f = \frac{2\pi - \frac{\alpha t_2^2}{2}}{t_2} \dots (3)$$

Validation is required due to the squaring step in the algebraic manipulation!!

Sub $\alpha = 5.79 \text{ rad/s}^2$ to (3): $\omega_f = \frac{2\pi - \frac{5.79(0.61^2)}{2}}{0.61} = \frac{2\pi - \frac{5.79(0.61^2)}{2}}{0.61} = \frac{2\pi - 1.077}{0.61} = 8.5 \text{ rad/s} > 0$

Sub $\alpha = 196.86 \text{ rad/s}^2$ to (3): $\omega_f = \frac{2\pi - \frac{196.86(0.61^2)}{2}}{0.61} = \frac{2\pi - \frac{196.86(0.61^2)}{2}}{0.61} = \frac{2\pi - 36.625}{0.61} = -49.74 \text{ rad/s} < 0$ (rejected !!)

(ii) how long did it take to make the first complete revolution?

Answer:

Sub $\alpha = 5.79 \text{ rad/s}^2$ to (1A): $t_1 = \sqrt{\frac{4\pi}{\alpha}} = \sqrt{\frac{4\pi}{5.79}} = 1.47 \text{ s}$

$$2\pi = 0 \times t_1 + \frac{1}{2} \alpha t_1^2 \dots (1)$$

$$\rightarrow 4\pi = \alpha t_1^2$$

$$\rightarrow t_1 = \sqrt{\frac{4\pi}{\alpha}} \dots (1A)$$

21

Rotational kinetic energy and moment of inertia

- Particle P has **kinetic energy** $KE = \frac{1}{2}mv^2 = \frac{1}{2}mr^2\omega^2$.

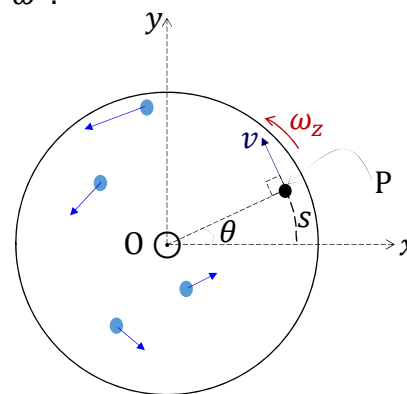
- The **rotational kinetic energy** of the rigid body rotating about a fixed axis with an angular speed ω ,

$$KE = \sum \frac{1}{2}mr^2\omega^2 = \frac{1}{2}\left(\sum mr^2\right)\omega^2 = \frac{1}{2}I\omega^2$$

where the **moment of inertia** of the rigid body about the **axis of rotation**,

$$I = \sum mr^2, \text{ or } I = \int r^2 dm.$$

I is obtained by multiplying the mass of each particle by the square of its **perpendicular distance** from the axis of rotation and adding these products.



22

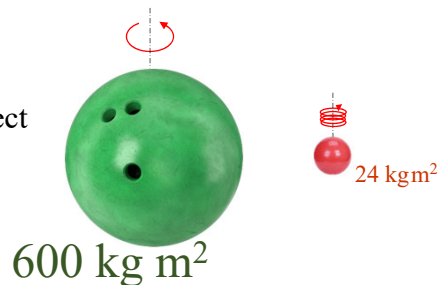
Mass versus Moment of Inertia

Mass in **kg** is the measurement of reluctance for an object to start a translational motion.

50 kg



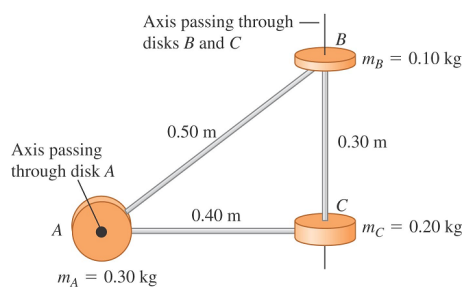
Moment of inertia in **kg m²** is the measurement of reluctance for an object to start a rotational motion.



23

Example 3. Moments of inertia about different rotation axis

A machine part consists of 3 disks linked by light-weight struts:



Its **moment of inertia** about an axis through the centres of disks B and C,

$$I_{BC} = m_A r_A^2 = 0.3 \times 0.4^2 = 0.048 \text{ kg m}^2.$$

Its **moment of inertia** about an axis through the centre of disk A, perpendicular to the plane of the diagram,

$$\begin{aligned} I_A &= m_B r_B^2 + m_C r_C^2 \\ &= 0.1 \times 0.5^2 + 0.2 \times 0.4^2 \\ &= 0.057 \text{ kg m}^2. \end{aligned}$$

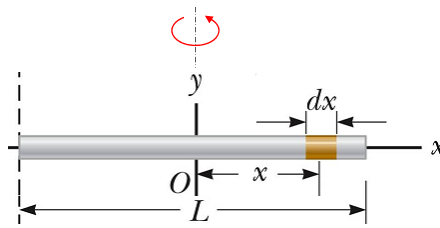
If the body rotates about the axis through A with **angular speed** $\omega = 4.0 \text{ rad/s}$, its **rotational kinetic energy** is

$$\begin{aligned} \frac{1}{2} I_A \omega^2 &= \frac{1}{2} (0.057) 4^2 \\ &= 0.456 \text{ J}. \end{aligned}$$

24

Example 4. The moment of inertia of a uniform rigid rod

Consider a uniform rigid rod of length L and mass M . We drill a small hole on its center.



The **moment of inertia** of the rod about an axis perpendicular to the rod and passing through its **center of mass** is

$$\begin{aligned}
 I_{cm} &= \int_{-\frac{L}{2}}^{\frac{L}{2}} x^2 dm = \int_{-\frac{L}{2}}^{\frac{L}{2}} x^2 \left(\frac{M}{L} dx \right) \\
 &= \frac{M}{L} \int_{-\frac{L}{2}}^{\frac{L}{2}} x^2 dx = \frac{M}{3L} [x^3]_{-\frac{L}{2}}^{\frac{L}{2}} = \frac{1}{12} ML^2.
 \end{aligned}$$

25

Problem. In a machine the rotation of a metal rod is not perpendicular to its axis of rotation. If the metal rod has mass M and length ℓ . Calculate the moment of inertia of the rod about an axis which is passing through its centre of mass and forming an angle θ to the rod.

Answer:

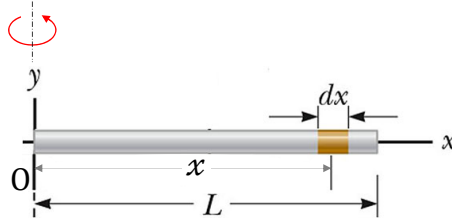
$$\begin{aligned}
 I &= \int r^2 dm = \int_{-\frac{\ell}{2}}^{\frac{\ell}{2}} (b \sin \theta)^2 \left(\frac{M}{\ell} db \right) \\
 &= \frac{M \sin^2 \theta}{\ell} \int_{-\frac{\ell}{2}}^{\frac{\ell}{2}} b^2 db = \frac{M \sin^2 \theta}{\ell} \left[\frac{b^3}{3} \right]_{-\frac{\ell}{2}}^{\frac{\ell}{2}} \\
 &= \frac{M \sin^2 \theta}{\ell} \left(\frac{\left(\frac{\ell}{2} \right)^3}{3} - \frac{\left(-\frac{\ell}{2} \right)^3}{3} \right) = \frac{M \sin^2 \theta}{\ell} \left(\frac{\ell^3}{8} \times \frac{2}{3} \right) = \boxed{\frac{M \ell^2 \sin^2 \theta}{12}}
 \end{aligned}$$

What if $\theta = 90^\circ$?

26

Example 5. The moment of inertia of a uniform rigid rod

Consider a uniform rigid rod of length L and mass M .



Instead of drilling the hole on the center, what if we drill the new hole just near the left end of the rod. The **moment of inertia** of the rod about the new axis perpendicular to the rod and passing through its left end is

$$I_{\text{New Hole}} = \int_0^L x^2 dm = \int_0^L x^2 \left(\frac{M}{L} dx \right) = \frac{M}{L} \int_0^L x^2 dx = \frac{M}{3L} [x^3]_0^L = \frac{1}{3} ML^2.$$

27

Example 5. The moment of inertia of a uniform rigid rod (cont.)

Remark 1 – the **parallel-axis theorem**.

We note that

$$I = \frac{1}{3} ML^2 = \frac{4}{12} ML^2 = \frac{1}{12} ML^2 + \frac{3}{12} ML^2 = I_{cm} + \frac{1}{4} ML^2 = I_{cm} + M \left(\frac{L}{2} \right)^2.$$

This is an example of the **parallel-axis theorem**. In general, for any rigid body of mass M , the **moment of inertia** about any axis parallel to an axis through its **centre of mass** but displaced from it by a distance d ,

$$I = I_{cm} + Md^2. \quad (\text{This formula can be proven.})$$

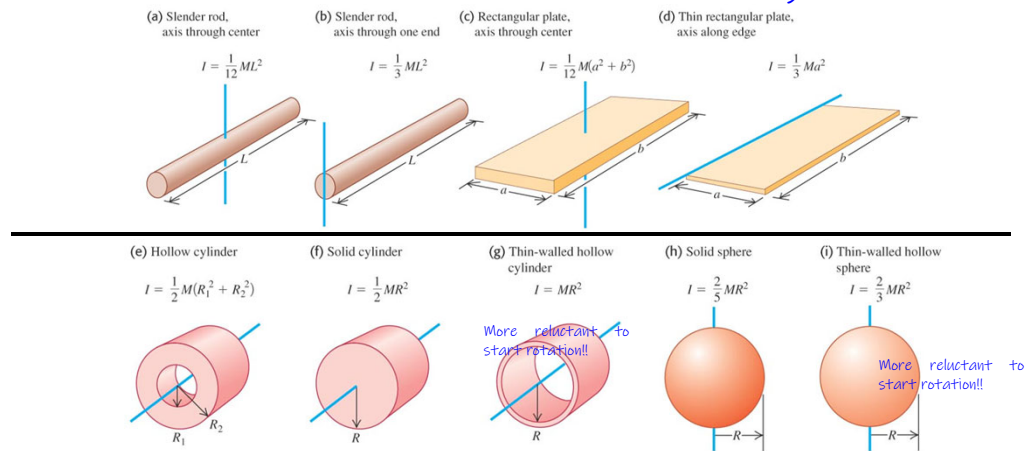
A rigid body has a lower moment of inertia about an axis through its centre of mass than about any other parallel axis.

28

Rotational kinetic energy and moment of inertia (cont.)

Remark 2. I depends on the body's mass and how it is distributed in space.

For a body with a given rotation axis and a given total mass, the greater the perpendicular distance from the axis to the particles that make up the body, the greater the moment of inertia. *(Hollow objects have larger I !!)*



29

Summary

Moment of Inertia:

(Resistance for Rotation)

$$I = \sum mr^2$$

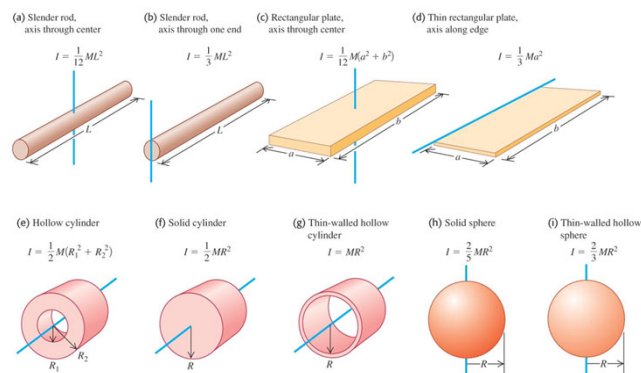
$$I = \int r^2 dm$$

Parallel Axis Theorem

$$I = I_{cm} + Md^2$$

$$KE_{\text{rotation}} = \frac{1}{2} I \omega^2$$

$$KE_{\text{translation}} = \frac{1}{2} mv^2$$



What is rotation?

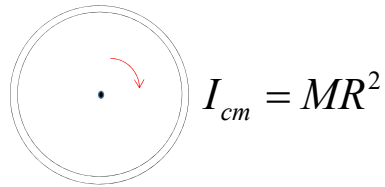
What is translation?

What is rolling?

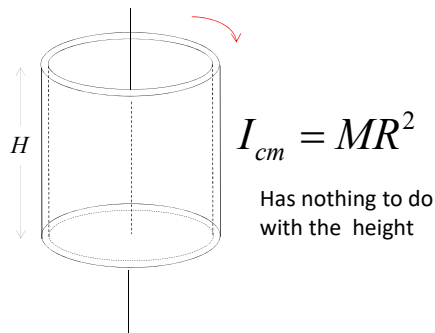
What is pure rolling?

30

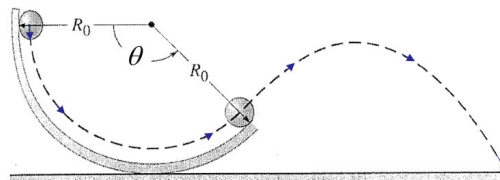
Moment of Inertia of a Ring of mass M and radius R :



Moment of Inertia of a Hollow Cylinder of mass M and radius R and Height H :



Problem. A uniform **solid ball** of mass m and radius r_0 **rolls** on the inside of a stationary track of radius R_0 without slipping as shown in the figure. The ball starts from rest at the vertical edge of the track as shown in the following figure. The moment of inertia for a solid ball of mass m and radius r is $\frac{2}{5}mr^2$. Assume that the track has negligible thickness, and the ball finally lands on the horizontal ground.



- (i) What will be the speed of the ball when it reaches the lowest point of the track?
- (ii) What will be the speed of the ball when it reaches the end of the track?
- (iii) If $r_0 = 1.5$ cm, $R_0 = 26.0$ cm and $\theta = 135^\circ$, what will be the horizontal distance from the lowest point of the track to the landed position of the ball on the ground?
- (iv) Explain how the results to (iii) will change (if at all) if the ball is replaced by a **spherical shell** of same mass and radius.

Rotation, Translation, Rolling

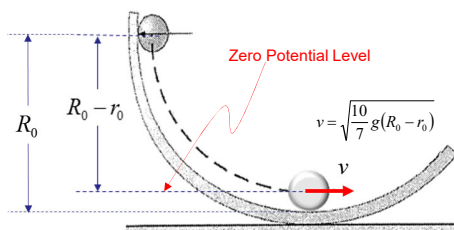


33

- (i) What will be the speed of the ball when it reaches the lowest point of the track?

Answer:

Let v be the speed of the ball at the lowest point of the track. We first apply the Conservation of Energy for the center of mass of the solid ball:



At the lowest point of the track, the ball has lost a height of $R_0 - r_0$.

This loss of potential energy is given to the linear speed **and** the angular speed of the ball at the bottom. So,

$$mg(R_0 - r_0) = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$v = r\omega \Rightarrow \omega = \frac{v}{r}$$

$$mg(R_0 - r_0) = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mr_0^2\right)\left(\frac{v}{r_0}\right)^2$$

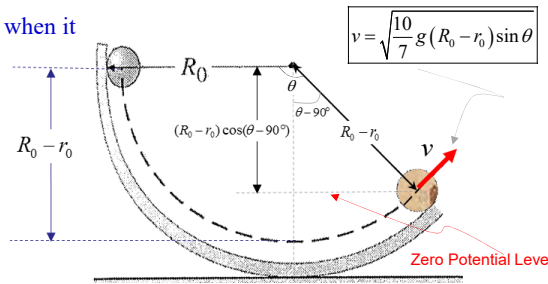
$$g(R_0 - r_0) = \frac{1}{2}v^2 + \frac{1}{5}v^2 = \frac{7}{10}v^2 \quad \text{So } v = \sqrt{\frac{10}{7}g(R_0 - r_0)} \text{ at the lowest point of the track.}$$

34

- (ii) What will be the speed of the ball when it reaches the end of the track?

Answer:

Let v be the speed of the ball at the end of the track. Again, we will apply the Conservation of Energy:



$$mg(R_0 - r_0)\cos(\theta - 90^\circ) = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \quad \text{but } \cos A = \cos(-A)$$

$$mg(R_0 - r_0)\cos(90^\circ - \theta) = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mr_0^2\right)\left(\frac{v}{r_0}\right)^2$$

$$g(R_0 - r_0)\sin \theta = \frac{1}{2}v^2 + \frac{1}{5}v^2 = \frac{7}{10}v^2$$

$$v = r\omega \rightarrow \omega = \frac{v}{r}$$

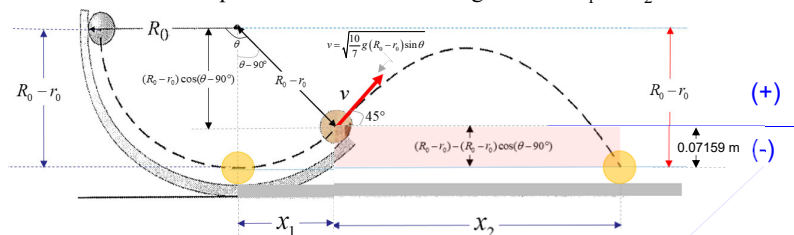
$$\boxed{\cos(90^\circ - \theta) = \sin \theta} \quad v = \sqrt{\frac{10}{7}g(R_0 - r_0)\sin \theta} \quad (\text{at the end of the track})$$

35

- (iii) If $r_0 = 1.5$ cm, $R_0 = 26.0$ cm and $\theta = 135^\circ$, what will be the horizontal distance from the lowest point of the track to the landed position of the ball on the ground?

Answer:

Refer to the following diagram. The horizontal distance from the lowest point of the track to the landed position of the ball on the ground is $x_1 + x_2$.



$$x_1 = (R_0 - r_0)\sin(\theta - 90^\circ) = (26 - 1.5)\sin(135^\circ - 90^\circ) = 17.3241 \text{ cm} = 0.1732 \text{ m}$$

The vertical distance from the end of the track down to the ground is

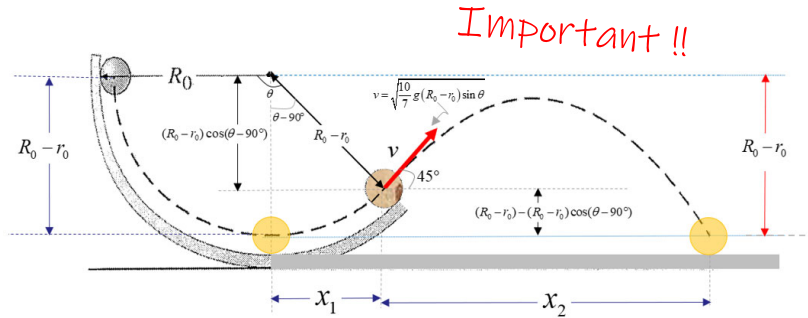
$$(R_0 - r_0) - [(R_0 - r_0)\cos(\theta - 90^\circ)] = (26 - 1.5) - (26 - 1.5)\cos(135^\circ - 90^\circ) = 7.1759 \text{ cm} = 0.07159 \text{ m}$$

To compute x_2 , we first compute the time required for the ball to travel from the end point of the track to the landed point on the ground based on the

vertical distance by the kinematic equation: $-0.07159 = (v\sin 45^\circ)t + \frac{1}{2}(-g)t^2$

$$\Delta y = v_y t + \frac{1}{2}at^2$$

36



x_2 **cannot** be computed
by the range formula: $R = \frac{v_i^2 \sin 2\theta}{g}$
because this portion of the trajectory
is not symmetrical !!

37

At the end of track,

$$v = \sqrt{\frac{10}{7} g (R_0 - r_0) \sin \theta}$$

$$= \sqrt{\frac{10}{7} \times 9.8 \times ((26 - 1.5) \times 10^{-2}) \times \sin 135^\circ} = \sqrt{2.4254} = 1.5574 \text{ m/s}$$

Sub v into vertical distance: $\Delta y = v_i t + \frac{1}{2} a t^2$
 $-0.07159 = (1.5574 \times \sin 45^\circ) t + \frac{1}{2} \times (-9.8) \times t^2$

$$4.9 t^2 - 1.101221 t - 0.07159 = 0$$

$$t = \frac{-(-1.101221) \pm \sqrt{(-1.101221)^2 - 4 \times 4.9 \times (-0.07159)}}{2 \times 4.9}$$

$$= \frac{1.101221 \pm \sqrt{1.212688 + 1.403164}}{9.8} = \frac{1.101221 \pm \sqrt{2.615852}}{9.8}$$

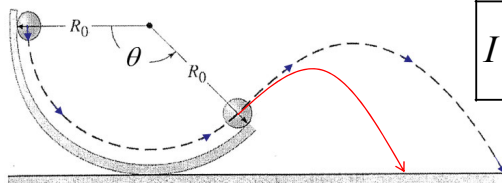
$$= \frac{1.101221 \pm 1.6174}{9.8} = 0.2774 \text{ s (take the positive value)}$$

$$x_2 = (v \cos 45^\circ) t = (1.5574 \text{ m/s} \times 0.7071) \times 0.2774 \text{ s} = 0.3055 \text{ m}$$

$$x_1 + x_2 = 0.1732 \text{ m} + 0.3055 \text{ m} = 0.4787 \text{ m} = \boxed{0.48 \text{ m}}$$

38

- (iv) Explain how the results to (iii) will change (if at all) if the ball is replaced by a **spherical shell** of same mass and radius.



$$I_{\text{Sphere}}^{\text{Solid}} = \frac{2}{5} MR^2$$

$$I_{\text{Sphere}}^{\text{Hollow}} = \frac{2}{3} MR^2$$

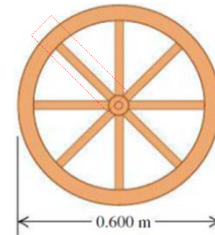
Answer:

The moment of inertia for a spherical **shell** is larger so it will require more energy to perform the rotation. As such the remaining energy to perform the translation is lesser. Therefore, the leaving translational speed of the ball at the end of the track will be slower so the horizontal distance travelled by the hollow sphere before it hits the ground **will be shorter**.

39

Question. A wagon wheel is constructed as shown in the figure. The radius of the wheel is 0.300 m, and the rim has mass 1.48 kg. Each of the eight spokes that lie along a diameter and are 0.300 m long and has mass 0.300 kg.

What is the moment of inertia of the wheel about an axis through its centre and perpendicular to the plane of the wheel?



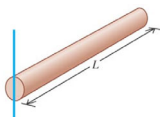
Answer:

The total moment of inertia is equal to the summation of the eight spokes (each regarded as a rod) and the rim (regarded as a thin cylinder) moment of inertia about the center of the wheel,

$$I_{\text{total}} = \left(\frac{1}{3} (0.3 \text{ kg}) (0.3 \text{ m})^2 \right) \times 8 + (1.48 \text{ kg}) (0.3 \text{ m})^2 = 0.21 \text{ kg m}^2$$

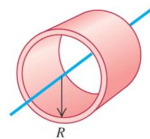
Slender rod,
axis through one end

$$I = \frac{1}{3} ML^2$$



Thin-walled hollow
cylinder

$$I = MR^2$$



40

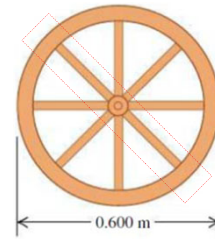
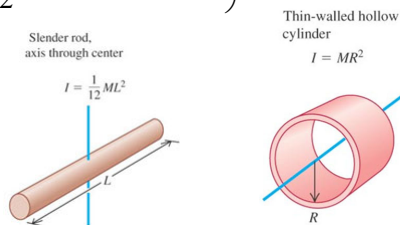
Question. A wagon wheel is constructed as shown in the figure. The radius of the wheel is 0.300 m, and the rim has mass 1.48 kg. Each of the eight spokes that lie along a diameter and are 0.300 m long and has mass 0.300 kg.

What is the moment of inertia of the wheel about an axis through its centre and perpendicular to the plane of the wheel?

Answer:

Or we can treat 2 spokes as one long bar with mass 0.6 kg, and length 0.6 m. Altogether there will be 4 long bars.

$$I_{\text{total}} = \left(\frac{1}{12} (0.6 \text{ kg}) (0.6 \text{ m})^2 \right) \times 4 + (1.48 \text{ kg}) (0.3 \text{ m})^2 = 0.21 \text{ kg m}^2$$



41

Question. The figure shows a hollow cylinder of uniform mass density ρ with length L , inner radius R_1 , and outer radius R_2 . Find its moment of inertia about its axis of symmetry. Express your answer in terms of the mass m of the hollow cylinder.

Answer:

The hollow cylinder can be broken down into many thin cylindrical films each of radius r and thickness dr and therefore having a moment of inertia of $dI = r^2 dm$, where $dm = \rho \times (2\pi r) \times L \times dr$. For the whole hollow cylinder,

$$I = \int dI = \int_{R_1}^{R_2} r^2 (\underbrace{\rho \times (2\pi r) L}_{dm} dr) = \int_{R_1}^{R_2} (2\pi \rho L r^3) dr$$

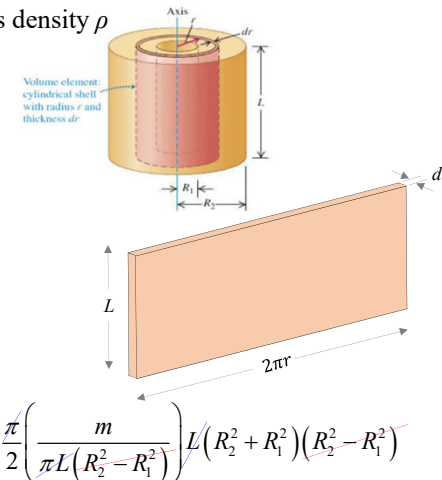
$$= 2\pi \rho L \left[\frac{r^4}{4} \right]_{R_1}^{R_2} = \frac{\pi}{2} \rho L (R_2^4 - R_1^4) \quad \dots (1)$$

If the **total mass** of such a hollow cylinder is m ,

$$\rho = \frac{m}{(\pi R_2^2)L - (\pi R_1^2)L} = \frac{m}{(\pi R_2^2 - \pi R_1^2)L} \quad \dots (2)$$

Sub (2) to (1):

$$I = \frac{\pi}{2} \left(\frac{m}{(\pi R_2^2 - \pi R_1^2)L} \right) L (R_2^4 - R_1^4)$$



$$= \frac{\pi}{2} \left(\frac{m}{\pi L (R_2^2 - R_1^2)} \right) L (R_2^2 + R_1^2) (R_2^2 - R_1^2)$$

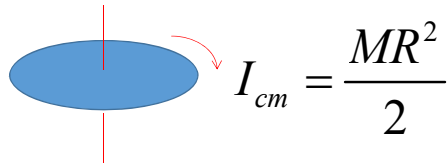
$$\Rightarrow I_{\text{Hollow Cylinder}} = \frac{1}{2} m (R_2^2 + R_1^2)$$

For solid cylinder, $R_1 \rightarrow 0$, $I_{\text{Solid Cylinder}} = \frac{1}{2} m R_2^2$

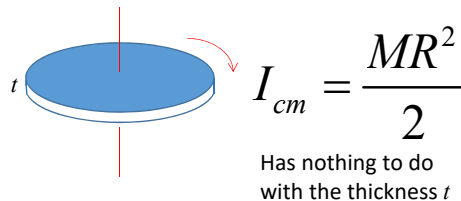
For thin cylindrical film, $R_1 \rightarrow R_2 = R$, $I_{\text{Thin Cylindrical Film}} = m R^2$

42

Moment of Inertia of a flat circular disc of mass M and radius R :

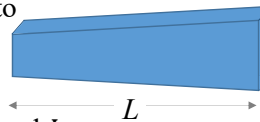


Moment of Inertia of a circular coin of mass M and radius R and thickness t



Problem. A slender rod with length L has a mass per unit length that varies with distance from the left end, where $x = 0$, according to

$$\frac{dm}{dx} = \beta x, \text{ where } \beta \text{ has units of kg/m}^2.$$



(i) Calculate the total mass M of the rod in terms of β and L .

Answer:

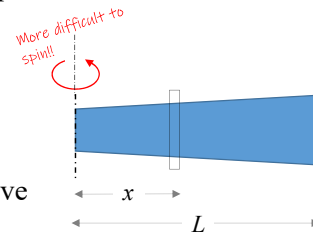
$$M = \int dm = \int_0^L \beta x dx = \beta \left[\frac{x^2}{2} \right]_0^L = \left[\frac{1}{2} \beta L^2 \right]$$

(ii) Calculate the moment of inertia I of the rod for an axis at the left end, perpendicular to the rod. Use the expression you derived in (i) to express I in terms of M and L . How does your result compare to that for a uniform rod? Explain.

Answer:

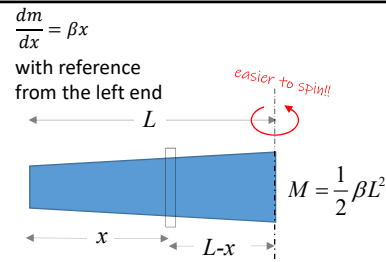
$$\begin{aligned} I &= \int x^2 dm = \int_0^L x^2 (\beta x dx) = \beta \int_0^L x^3 dx = \beta \left[\frac{x^4}{4} \right]_0^L \\ &= \frac{\beta}{4} L^4 = \frac{1}{2} \left(\frac{1}{2} \beta L^2 \right) L^2 = \frac{1}{2} M L^2 > \frac{1}{3} M L^2 \end{aligned}$$

I is greater than that for a uniform rod since the more massive parts of the rod are further away from the axis of rotation.



- (iii) Repeat (ii) for an axis at the right end of the rod. How do the results for (ii) and (iii) compare? Explain.

Answer:



$$\begin{aligned}
 I &= \int (L-x)^2 dm = \int_0^L (L-x)^2 (\beta x dx) = \beta \int_0^L (x^3 - 2Lx^2 + L^2x) dx \\
 &= \beta \left[\frac{1}{4}x^4 - \frac{2}{3}Lx^3 + \frac{1}{2}L^2x^2 \right]_0^L = \beta \left(\frac{1}{4} - \frac{2}{3} + \frac{1}{2} \right) L^4 = \beta \left(\frac{3-8+6}{12} \right) L^4 \\
 &= \frac{1}{12} \beta L^4 = \frac{1}{6} \left(\frac{1}{2} \beta L^2 \right) L^2 = \frac{1}{6} ML^2 < \frac{1}{3} ML^2
 \end{aligned}$$

I is smaller than that for a uniform rod since the more massive parts of the rod are now closer to the axis of rotation.

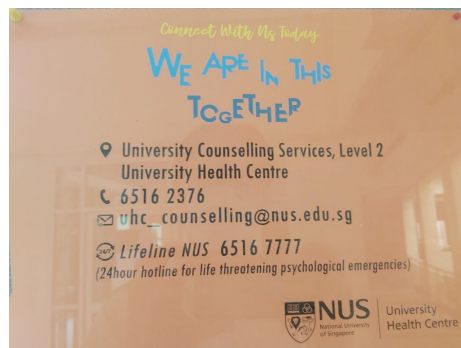
45

Helplines to de-stress

We are not superman/supergirl. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019

46